Data-Driven Sales and Inventory Optimization for a Rural Kirana Store

A Final report for the BDM capstone Project

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1. Executive Summary

Bhavana Provision General Store is a rural B2C kirana shop located in Pimpri Khurd, Taluka Mangrulpir, District Washim, Maharashtra, operated by Mr. Sunil Surve since 2015. The store serves daily grocery and household needs of nearby villages and deals mainly in essential commodities such as rice, wheat flour, pulses, cooking oil, sugar, biscuits, and other packaged goods. Despite steady sales and a strong customer base, the shop faces key challenges including high dependency on credit transactions, inefficient inventory management, and lack of a structured system for tracking business performance. These issues lead to cash flow imbalance, delayed restocking of fast-moving products, and difficulty in monitoring financial activities.

The dataset used for this analysis consists of 6465 transaction records containing variables such as Date, Item Name, Quantity, Price per Unit, Total Amount, Payment Mode, Amount Paid, and Pending Dues. The total sales recorded are approximately ₹6,34,657, with an average transaction value of ₹98.17 and a median value of ₹50, reflecting frequent small purchases typical of rural kirana stores. Out of total sales, ₹3,76,798 has been received while ₹2,57,859 remains pending, indicating a significant portion of credit transactions. Analytical techniques such as descriptive statistics, product-wise aggregation, Pivot Tables, and data visualization charts were used to analyze sales trends, customer behaviour, and financial patterns.

The analysis reveals that a small group of essential products contributes a major share of total sales and revenue, with items like cooking oil, pulses, and staple goods acting as primary revenue drivers due to their high and consistent demand. These products are fast-moving and frequently purchased, ensuring stable sales. In contrast, certain packaged and non-essential items have lower sales frequency but offer relatively higher profit margins per unit. The study also identifies that nearly 40% of total sales value remains in pending dues, indicating strong reliance on credit transactions. Additionally, variations in demand patterns and absence of structured inventory classification lead to stock imbalances, where fast-moving items face stock-outs and slow-moving items remain unsold.

Based on these findings, several recommendations are proposed to improve business performance. These include reducing credit dependency through better tracking of dues, maintaining higher stock levels for fast-moving essential products, minimizing investment in slow-moving items, and implementing a simple digital system such as Excel for monitoring sales and profit. Introducing inventory classification methods like ABC analysis can further improve stock management. These improvements can enhance cash flow, reduce inefficiencies, and support better decision-making. Overall, by adopting a data-driven approach, the store can improve financial control, optimize inventory, and achieve sustainable long-term growth.

2. Detailed Explanation of Analysis Process

The data for this project was collected from Bhavana Provision General Store, a rural kirana shop located in Pimpri Khurd, Taluka Mangrulpir, District Washim, Maharashtra. The shop maintains most of its transaction records manually in notebooks and informal registers. For the purpose of this study, these handwritten records were carefully digitized and converted into a structured dataset using Microsoft Excel. The final dataset consists of 6465 transaction records, capturing real sales activities over a defined time period. Each transaction includes important details such as product name, quantity sold, unit price, total amount, payment mode, amount paid, and pending dues. This data reflects actual business operations and helps in understanding customer purchasing behavior and financial performance of the shop.

2.1 Data Collection and Digitization

The initial step involved collecting raw data from daily shop records, including sales entries and customer credit details. Since the data was originally maintained in handwritten format, it required careful digitization to convert it into a usable form. Each transaction was entered into Excel with structured columns such as Date, Item Name, Quantity, Price per Unit, Total Amount, Payment Mode, Amount Paid, and Pending Dues. The aim of this step was to ensure that all real-world transactions were accurately represented in a digital format, allowing further analysis to be performed efficiently.

2.2 Data Cleaning and Structuring

After data entry, the dataset underwent a thorough cleaning and preprocessing process to ensure accuracy and consistency. Inconsistent product names were standardized to avoid duplication for example, variations of the same product were grouped under a single name. Missing or incorrect values in numeric fields such as Quantity, Price, and Amount were corrected. All numerical columns were converted into appropriate formats to support calculations.

Additional derived variables were created to enhance analysis. These include calculated fields such as Total Sales, Estimated Cost, Profit, and Month. This helped in performing time-based analysis and identifying trends across different periods. The cleaning process ensured that the dataset was reliable, consistent, and suitable for statistical and visual analysis.

2.3 Profit and Sales Analysis

Profit and sales analysis was conducted to understand which products contribute the most to revenue and profitability. It was observed that essential products such as Rice, Wheat Flour, and Cooking Oil generate a significant portion of total sales due to their high demand and frequent purchase. These items act as primary revenue drivers for the store.

At the same time, certain products such as Biscuits, Spices, and Packaged Items show relatively higher profit margins despite lower sales volume. This indicates that while staple goods drive volume, packaged goods contribute better margins per unit. This analysis helps distinguish between high-volume and high-margin products, which is important for optimizing overall profit.

2.4 Inventory Movement and Product Classification

To understand stock movement, product-level sales patterns were analyzed. Fast-moving products such as staple food items show consistent demand and require frequent restocking. In contrast, some products move slowly and remain in stock for longer periods, leading to inefficient use of storage and capital.

An informal classification of products was performed based on their contribution to total sales. High-demand items were categorized as primary products, while moderate and low-demand items were grouped accordingly. This classification highlights the need for prioritizing inventory based on product performance and demand frequency.

2.5 Customer Payment Behaviour and Cash Flow Analysis

Customer payment behaviour was analyzed using the Payment Mode, Amount Paid, and Pending Dues fields. The analysis shows that approximately 40% of transactions were made on credit, indicating a significant reliance on delayed payments. This creates a gap between recorded sales and actual cash inflow.

Additionally, a portion of total sales remains unpaid in the form of pending dues, which affects the store's ability to maintain steady cash flow and reinvest in inventory. Since the shop relies on manual tracking of credit transactions, monitoring outstanding payments becomes difficult. This highlights the need for implementing a structured credit management system.

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2.6 Demand Pattern Analysis and Forecasting

The dataset was further analyzed to identify changes in product demand over time. Essential products show consistent demand across the study period, while certain packaged items exhibit fluctuations based on consumption patterns and external factors.

Basic trend analysis indicates that regularly used household items maintain stable sales, whereas non-essential products show variability. Based on observed patterns, it is expected that essential goods will continue to perform steadily in the future, while demand for non-essential items may fluctuate.

This analysis emphasizes the need for demand-based stock planning and forecasting to avoid both overstocking and stock shortages.

3. Results and Findings

3.1 Credit Impact and Cash Flow Imbalance

This section evaluates the impact of customer credit transactions on the cash flow of Bhavana Provision General Store using total sales, amount received, and pending dues data.

$$\text{Credit Ratio} = \frac{\text{Pending Dues}}{\text{Total Sales}} \times 100$$

It measures how much of the total revenue is locked in credit instead of being available as liquid cash.

Key Observations:

- High Credit Level (~40.6%) **:** A significant portion of total sales ₹2,57,859 out of ₹6,34,657 remains unpaid, indicating heavy reliance on credit transactions.

- Cash Flow Pressure **:** Only ₹3,76,798 is realized as actual cash inflow, which limits immediate reinvestment capability and affects liquidity.
- Delayed Recovery Risk **:** High pending dues increase the risk of delayed or incomplete payments from customers.
- Operational Dependency **:** The shop heavily depends on customer trust rather than structured credit tracking systems.

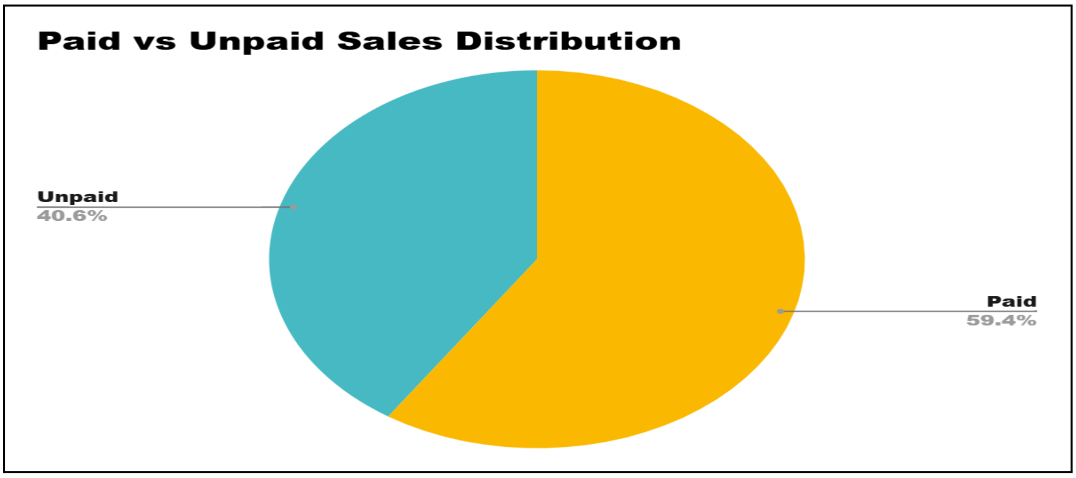

Fig1: Distribution of Cash vs Credit Sales

Detailed Analysis:

The credit analysis clearly shows that a large portion of revenue is tied up in pending payments, reducing liquidity. This restricts the shop owner's ability to purchase stock on time, especially fast-moving items. Since credit tracking is done manually, it becomes difficult to monitor outstanding balances efficiently.

The gap between total sales and actual cash inflow highlights a significant cash flow imbalance. This situation increases financial risk and reduces operational efficiency. Implementing a structured credit management system is essential to ensure timely recovery of dues and maintain a stable cash flow.

3.2 Sales Trend Analysis (Monthly Performance)

This section analyzes sales variation across time to identify patterns in revenue generation.

Key Observations:

- Fluctuating Sales Trend : Sales show variation across months, with a peak in November and a decline in January, indicating changing customer demand patterns.

- Stable Core Demand : Essential products maintain relatively stable contribution to total sales despite minor fluctuations.
- Demand Sensitivity **:** Non-essential and packaged goods show higher fluctuations compared to staple items.

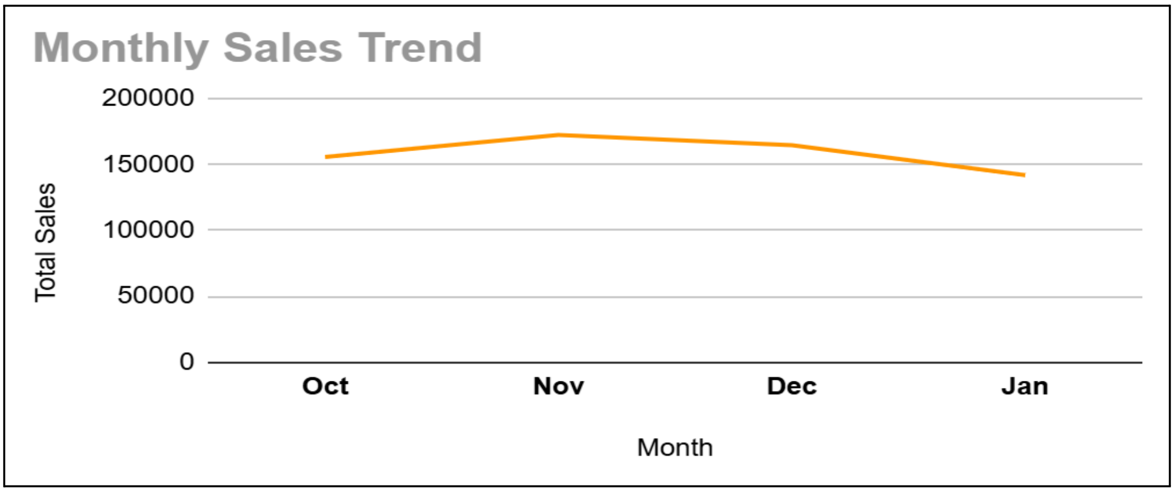

Fig2: Monthly Sales Trend showing variation in total revenue

Detailed Analysis:

The monthly sales trend indicates moderate fluctuations over time, with the highest sales observed in November followed by a gradual decline in December and January. This pattern reflects variations in customer purchasing behavior, possibly influenced by seasonal and consumption factors.

Essential items such as rice, flour, and oil maintain consistent demand, ensuring a stable base of revenue. In contrast, non-essential and packaged goods exhibit higher variability, indicating demand sensitivity.

Overall, the sales trend suggests a necessity-driven market where core products drive consistent revenue, while other products contribute to fluctuations. This highlights the importance of balancing inventory between stable and variable demand items for efficient stock management.

This emphasizes the need for aligning inventory decisions with demand trends to maintain operational efficiency.

3.3 Product Profitability and Margin Performance

This section evaluates product-level profitability to identify high-margin and low-margin items.

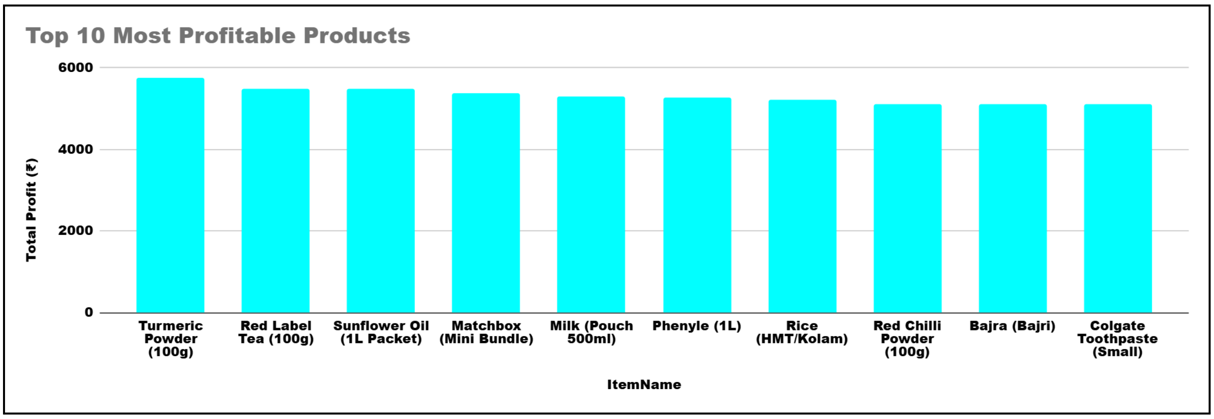

Fig 3: Top 10 Most Profitable Products

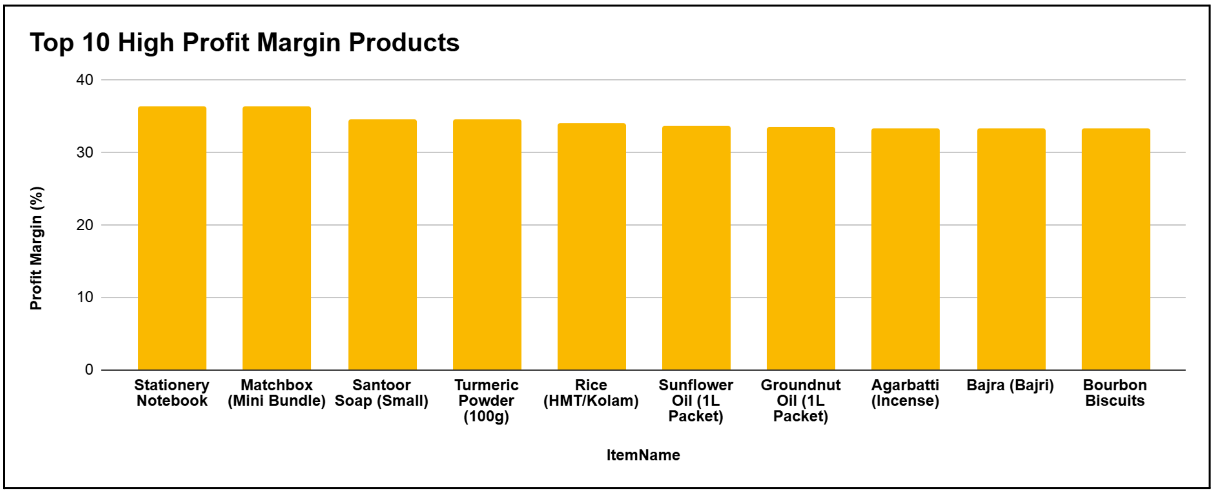

Fig 4: Top 10 Products by Profit Margin (%)

Key Observations:

- High Profit vs High Margin Difference : Products such as Turmeric Powder (100g), Red Label Tea (100g), and Sunflower Oil (1L Packet) generate higher total profit due to strong sales volume, while products like Stationery Notebook and Matchbox (Mini Bundle) show higher profit margins but do not necessarily contribute the highest total profit.
- Consistent Margin Range : The top 10 products maintain profit margins within a narrow range of approximately 32% to 36%, indicating a stable pricing and cost structure across products.
- Volume-Driven Profitability : Essential and frequently purchased products (e.g., oil, pulses, and daily-use items) contribute more to total profit due to higher sales volume, even if their margins are moderate.

- Balanced Profit Distribution : Profit is not dependent on a single product; instead, multiple products contribute to overall profitability, reducing business risk.

Detailed Analysis:

The analysis highlights a clear distinction between total profit and profit margin. While some products generate higher profit margins, they do not always result in the highest total profit due to lower sales volume. On the other hand, essential and fast-moving products contribute significantly to total profit because of their high demand and frequent purchase.

The relatively consistent margin range (32%–36%) suggests that the store follows a uniform pricing strategy across different product categories. This stability indicates effective cost control and standardized markup practices.

Overall, the store's profitability is driven by a combination of high-volume essential products and moderate-to-high margin items. To maximize profit, the business should focus on maintaining stock availability for high-volume products while also promoting high-margin items to improve overall profitability.

3.4 Product Contribution and Inventory Classification

This section classifies products based on their contribution to total sales and demand frequency.

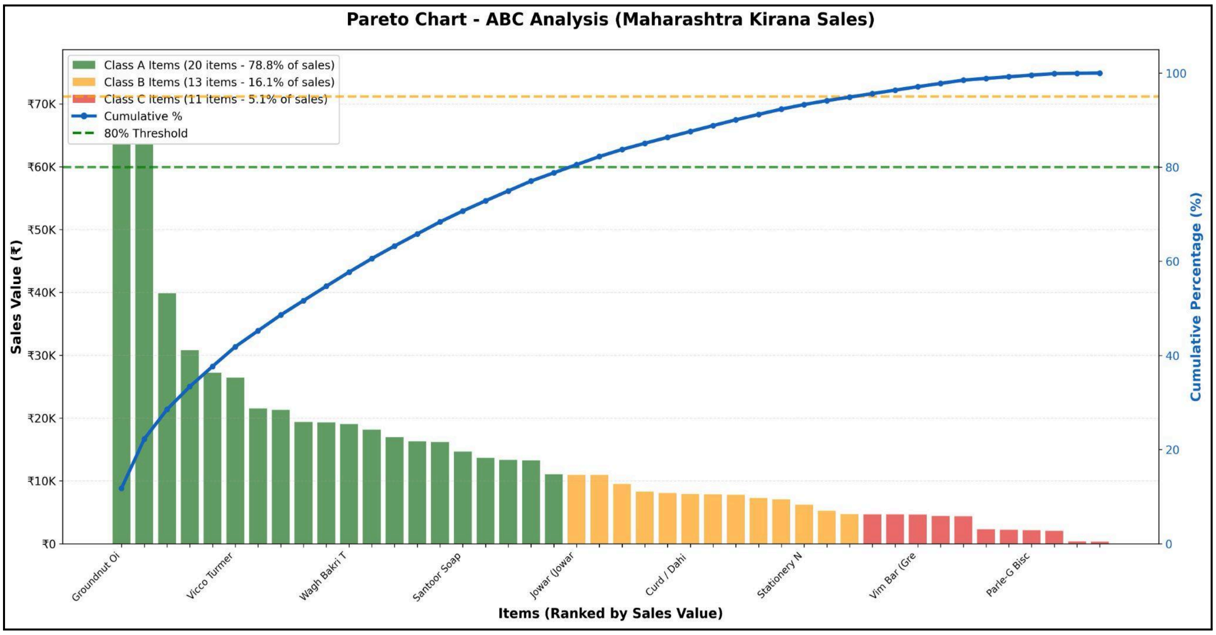

Fig5:Item-wise sales ranking with cumulative percentage

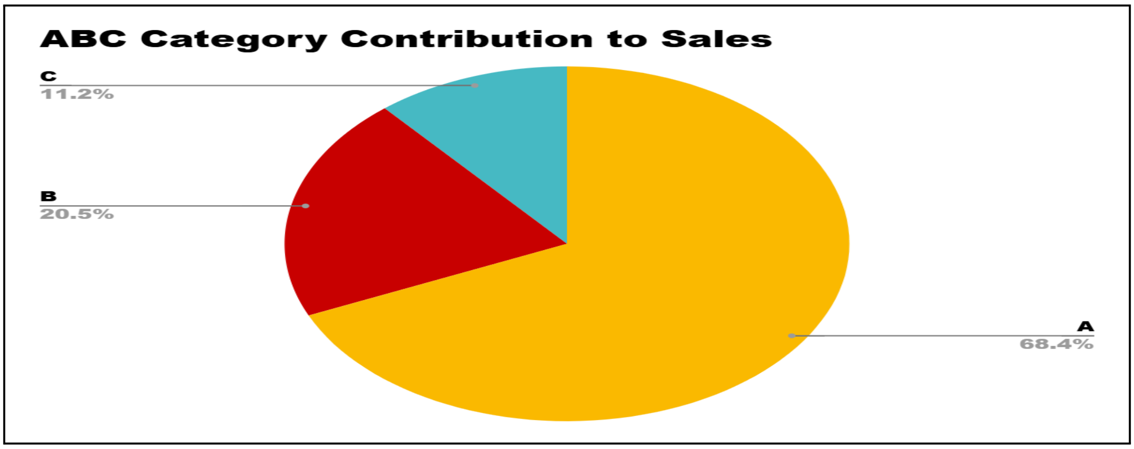

Fig6: ABC Category Contribution to Total Sales

Key Observations:

- High Contribution Products : Category A items contribute the largest share of total revenue (approximately 68.4%), indicating that a small number of products drive most of the sales.
- Moderate Contribution Products : Category B items contribute around 20.5% of total revenue, showing stable and consistent demand across mid-level products.
- Low Contribution Products : Category C items contribute only about 11.2% despite being higher in number, indicating low-demand or slow-moving inventory.
- Revenue Concentration : A significant portion of revenue is concentrated in a limited number of high-performing products, highlighting the importance of prioritization.

Detailed Analysis:

The ABC classification clearly demonstrates that the store's revenue is heavily dependent on Category A products, which contribute the majority of total sales. These products are typically high-demand and fast-moving, requiring efficient stock management to prevent stock-outs.

Category B products provide moderate and stable contributions, acting as a supporting layer to overall revenue. Maintaining optimal inventory levels for these items ensures consistent sales without excessive stock accumulation.

In contrast, Category C products contribute minimally to revenue while occupying storage space. These items represent slow-moving inventory and may lead to increased holding costs if not managed properly.

Overall, the analysis suggests that inventory planning should focus on ensuring availability of Category A products, maintaining balanced stock for Category B items, and reducing excess inventory of Category C products to improve overall efficiency and profitability.

3.5 Product-wise Seasonal Demand Analysis (Agarbatti vs Vicco Turmeric Cream)

This section analyzes demand behavior and predicts future trends based on observed patterns.

Key Observations:

- Seasonal Sales Variation : Sales show fluctuation across months, with an increase from October to November, followed by a dip in December and a sharp rise in January for Vicco Turmeric Cream.
- Product-Level Demand Difference : Vicco Turmeric Cream shows significantly higher sales compared to Agarbatti, indicating stronger and more consistent demand.
- Fluctuating Demand for Agarbatti : Agarbatti (Incense) shows a gradual decline from October to December, with a slight recovery in January, indicating irregular or occasion-based demand.
- Strong Growth in January : Vicco Turmeric Cream experiences a notable increase in January, suggesting higher demand possibly due to seasonal or usage-related factors.

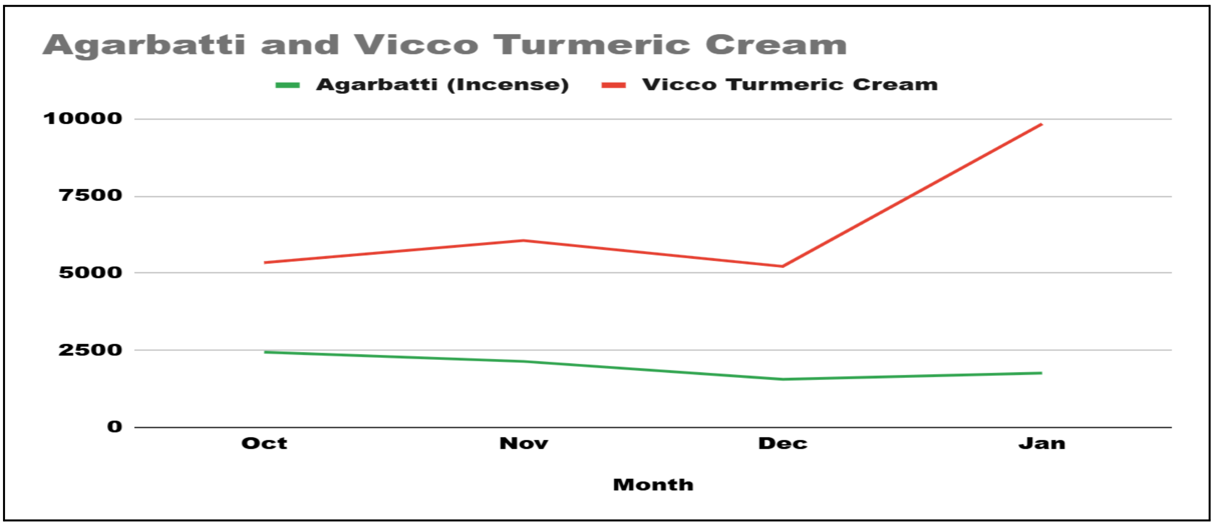

Fig7: Seasonal Demand Pattern of Agarbatti and Vicco Turmeric cream

Detailed Analysis:

The seasonal demand pattern indicates that different products exhibit distinct sales behavior over time. Vicco Turmeric Cream shows a generally upward trend, with a significant increase in January, suggesting consistent and growing demand. This may be influenced by factors such as increased usage during winter months or higher consumer preference for personal care products.

In contrast, Agarbatti(Incense) displays a declining trend from October to December, followed by a slight recovery in January. This indicates that its demand may be occasional or linked to specific events rather than consistent daily consumption.

The comparison highlights that essential or frequently used products tend to maintain or increase demand over time, while non-essential or occasion-based products show fluctuations.

From an inventory planning perspective, the store should ensure adequate stock of high-demand products like Vicco Turmeric Cream, especially before peak months. At the same time, cautious stocking of products like Agarbatti is recommended to avoid overstocking and reduce holding costs.

Overall, understanding product-specific demand variations enables better forecasting, optimized inventory management, and improved decision-making for future sales planning.

3.6 Customer Behaviour and Purchase Patterns

This section examines customer buying behavior based on transaction data.

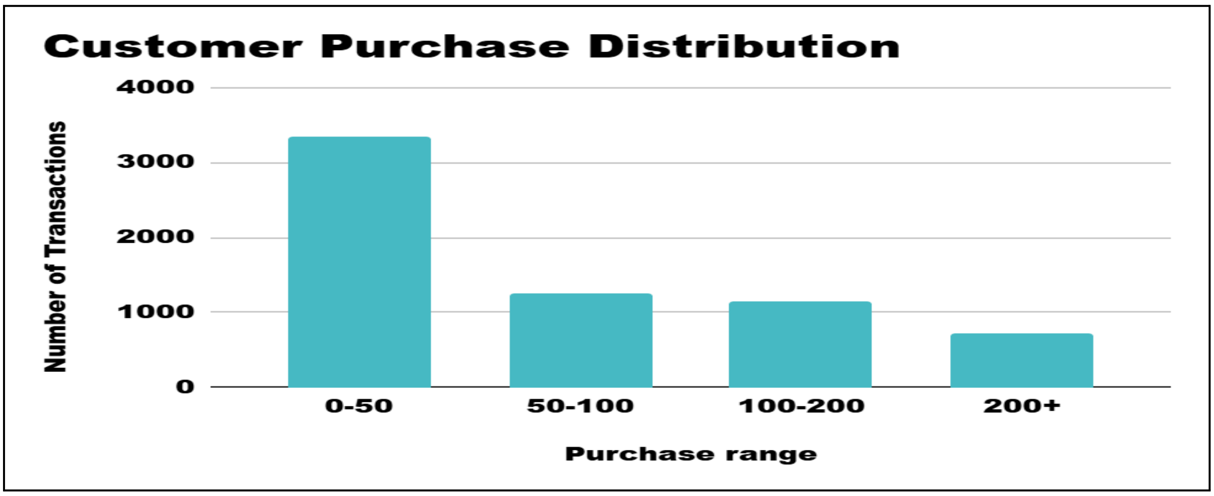

Fig8: Distribution of Customer Purchase Values

Key Observations:

- Dominance of Small Transactions : The majority of purchases fall in the ₹0–50 range, indicating that customers frequently make low-value transactions.

- Moderate Mid-Range Purchases : Transactions in the ₹50–100 and ₹100–200 ranges are moderate, showing occasional bulk or planned purchases.
- Low High-Value Purchases : Very few transactions fall in the ₹200+ category, indicating limited high-value spending.
- Frequent Visit Behavior : The pattern suggests that customers prefer making multiple small purchases instead of fewer large transactions.

Detailed Analysis:

The purchase distribution indicates that the store operates in a low-ticket-size environment where most customers prefer small and frequent purchases. The high concentration of transactions in the ₹0–50 range suggests daily or routine buying behavior, which is typical in local kirana stores.

Mid-range transactions ₹50–200 occur less frequently and likely represent planned purchases or slightly higher consumption needs. However, high-value transactions ₹200+ are minimal, indicating that customers rarely engage in bulk buying.

This pattern highlights the importance of maintaining consistent stock availability for frequently purchased low-cost items. It also suggests that strategies such as combo offers or discounts could be used to encourage higher-value purchases.

Overall, the customer behavior reflects a necessity-driven purchasing pattern with an emphasis on affordability and frequency rather than bulk buying.

3.7 Top Performing Product

This section highlights the most impactful product categories based on sales performance.

Key Observations:

- Top Revenue Contributors : Products such as Groundnut Oil 1L Packet and Sunflower Oil 1L Packet generate the highest sales revenue, making them the most impactful items.
- Staples Lead Sales : Essential items like pulses Tur Dal, flour-related products, and cooking oils dominate the top-selling list, indicating consistent demand.
- Mid-Level Contributors : Products like Wheel Powder, Red Label Tea, and Vicco Turmeric Cream contribute moderately to total revenue.

- Diversified Product Mix : The top 10 products belong to different categories such as staples, household goods, and personal care, ensuring balanced revenue sources.

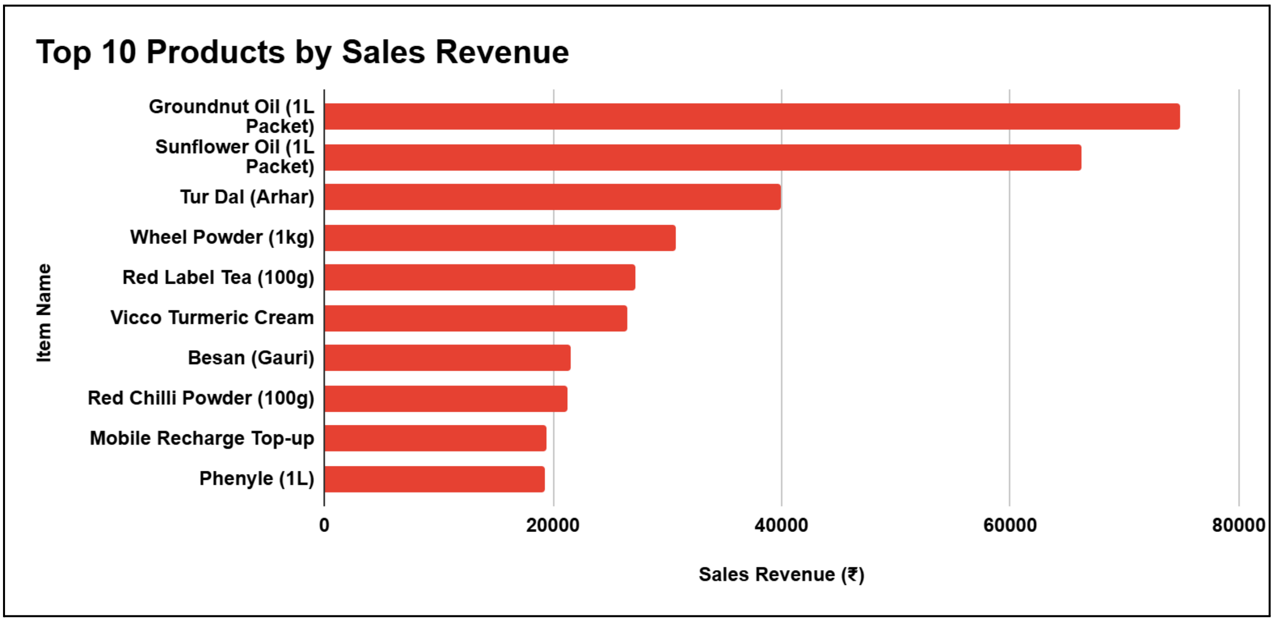

Fig9: Top 10 Products Contributing to Total Sales Revenue

Detailed Analysis:

The analysis shows that the store's revenue is largely driven by a few high-performing products, particularly essential grocery items like edible oils and pulses. These products consistently generate high sales due to their necessity in daily consumption.

Mid-level products contribute steadily but do not match the impact of top-performing items. Their presence in the top 10 indicates their importance in maintaining overall sales stability.

The diversity of product categories among top performers reduces dependency on a single category and helps in spreading risk across multiple segments.

Overall, the store should prioritize maintaining stock availability for these top-performing products while also promoting mid-level items to further enhance revenue. Strategic focus on high-demand essentials can ensure sustained business performance and growth.

This highlights the importance of prioritizing essential products that ensure consistent revenue while strategically promoting mid-level products to enhance overall profitability.

By focusing on both high-demand and moderate-performing items, the store can achieve a balance between stability and growth in sales performance.

3.8 Sales Forecasting and Comparison Analysis

This section evaluates future sales trends using simple forecasting techniques and compares forecasted values with actual sales data to assess expected demand patterns.

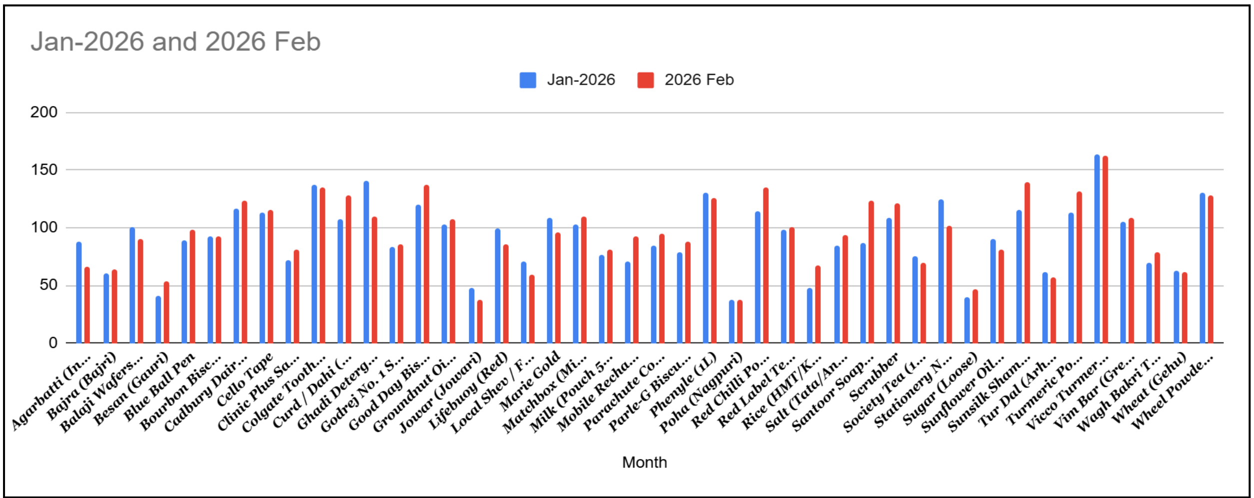

Fig10: Forecasted Sales and Comparison between Actual and Forecasted Values

Key Observations:

- Stable Core Demand : Daily-use products such as dairy items, biscuits, and basic grocery goods show consistent sales between January and February, indicating steady household consumption patterns.
- Growth in Selected Products : Certain items like personal care products and packaged goods show a noticeable increase in February compared to January, suggesting rising demand in hygiene and convenience-based categories.
- Decline in Few Categories : Some products exhibit a slight decrease in forecasted values, indicating reduced short-term demand or variability in customer preferences.
- Overall Similar Trend Pattern : The forecasted values closely follow the actual sales pattern of January, indicating that historical data is a reliable indicator for short-term demand prediction.

Detailed Analysis:

The forecasting analysis highlights that most products maintain a similar sales pattern in the upcoming period, especially essential and frequently purchased goods. This reflects a stable demand environment where customer buying behavior does not change drastically in the short term. At the same time, variations across certain products indicate that demand is not uniform. Some items experience growth due to changing consumption needs, while others show slight decline, possibly due to reduced necessity or seasonal effects.

The comparison between actual (January 2026) and forecasted (February 2026) sales suggests that demand trends remain predictable for most categories. This allows the store to make informed decisions regarding stock planning and purchasing frequency.

From a business perspective, this analysis helps in:

- Maintaining sufficient stock for consistently demanded products
- Identifying products with increasing demand for better availability
- Avoiding overstocking of slow-moving or declining items

Overall, sales forecasting supports better inventory planning, reduces uncertainty, and improves operational efficiency by enabling data-driven decision-making.

4. Interpretation of Results and Recommendations

Based on the analysis of Bhavana Provision Store's sales data, we found several problems. These include too much credit being given, poor inventory management, and lack of proper tracking. Here are solutions to improve cash flow and inventory efficiency.

4. Interpretation of Results and Recommendations

Based on the analysis of Bhavana Provision Store’s sales data, some major problems were identified. These include high use of credit, poor inventory management, and lack of proper tracking systems. The following suggestions can help improve cash flow and inventory efficiency.

4.1 Reduce Credit Dependency and Improve Cash Flow

Problem: Around 40–43% of total sales are on credit. This means the store does not get cash immediately, which makes it difficult to buy new stock.

Solution:

- Maintain a simple customer ledger with name, credit amount, and pending dues
- Set credit limits for regular customers
- Give small discounts for cash payments
- Follow up with customers every week to collect dues

4.2 Improve Inventory Planning for Fast-Moving Products

Problem: Important items like oil and dal go out of stock frequently, which leads to loss of sales.

Solution:

- Keep at least 1–2 weeks of stock for fast-moving items
- Check stock levels regularly
- Refill items before they are finished

4.3 Reduce Investment in Slow-Moving Products

Problem: Some products like Blue Ball Pen and Cello Tape do not sell much and occupy space.

Solution:

- Buy less quantity of slow-moving items
- Use discounts or combo offers to sell old stock
- Replace them with better-selling products

4.4 Implement ABC Inventory Classification

Problem: All products are treated the same, even though some sell more than others.

Solution:

- Divide products into three categories:
    - **A (Fast-moving):** 68.4% of sales
    - **B (Moderate):** 20.5% of sales
    - **C (Slow-moving):** 11.2% of sales

4.5 Inventory Movement Analysis

Problem: Products are not classified based on their sales performance. Fast-moving items like Groundnut Oil, Sunflower Oil, Tur Dal and slow-moving items like Blue Ball Pen, Cello Tape, Clinic Plus are managed the same way, leading to inefficient inventory allocation and missed opportunities for optimization.

Solution:

Category A (Fast-Moving Items): Products like Groundnut Oil, Sunflower Oil, Tur Dal, and Red Label Tea sell daily and generate high revenue. These items are very important for the business.

- Always keep enough stock available
- Maintain 2–3 weeks of buffer inventory
- Check stock levels every week

- Never let these items run out

Category B (Moderate-Moving Items): Items like Rice, Milk, Salt, and Agarbatti have stable and consistent demand.

- Review stock once a month
- Stock according to actual demand
- Use monthly reorder cycles

Category C (Slow-Moving Items): Products like Blue Ball Pen, Cello Tape, and Clinic Plus Sachet have very low sales.

- Buy only when customers ask for them
- Avoid unnecessary spending and storage space waste
- Consider replacing with faster-selling products

Refer to the following graphs for Fast-Moving vs Slow-Moving Items Sales comparison:

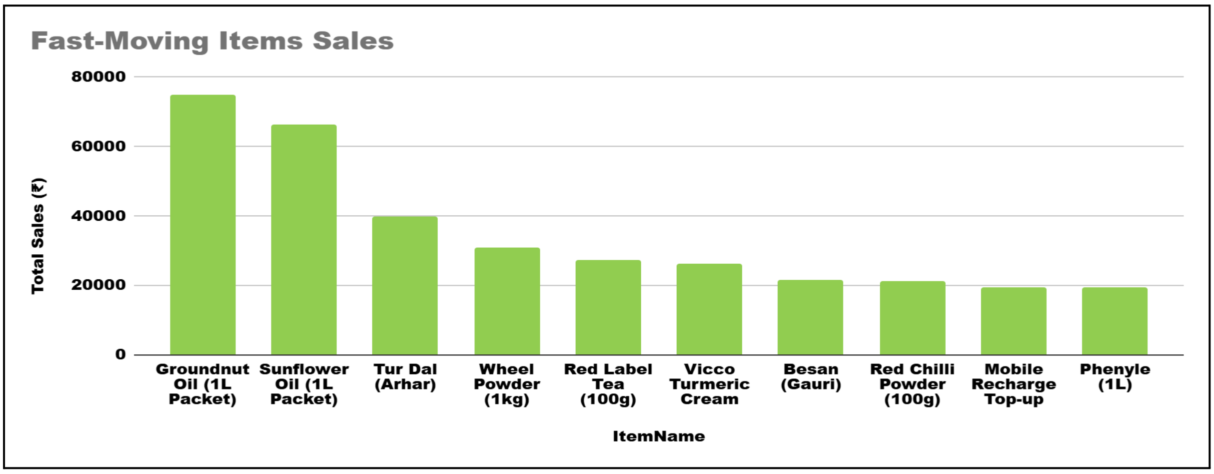

Fig11.Fast Moving Items

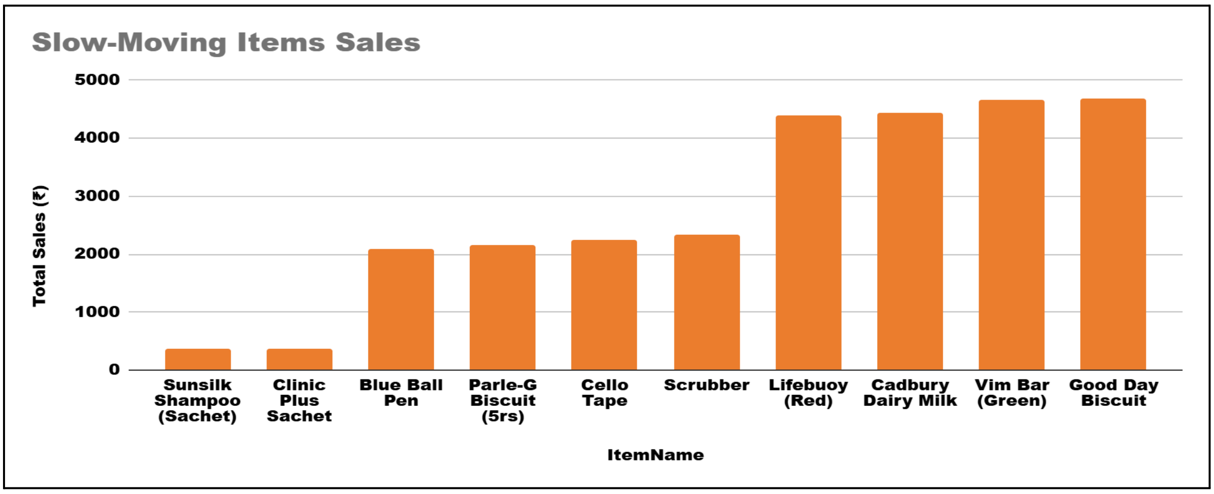

Fig12.Slow Moving Items

4.6 Strengthen Credit Monitoring System

Problem: Manual tracking of credit is difficult and payments are often delayed.

Solution:

- Use Excel to track customer dues
- Check pending payments weekly
- Send reminders to customers
- Avoid giving credit to customers who delay payments

4.7 Create a Sales Dashboard

Problem: There is no proper system to track sales and profit.

Solution:

- Create a simple dashboard using Excel Pivot Tables
- Track daily and monthly sales
- Identify high-profit products
- Use charts to understand trends easily

4.8 Use Historical Data for Buying Decisions

Problem: Stock is purchased based on guesswork.

Solution:

- Use past sales data to predict demand
- Buy more of high-demand items
- Reduce purchase of low-demand items
- Consider seasonal trends

4.9 Improve Product Placement

Problem: Some products are not visible to customers.

Solution:

- Place popular items near the counter
- Keep products at eye level
- Bundle slow-moving items with popular ones

4.10 Maintain Balance Between Volume and Profit

Problem: The store depends mainly on essential items with low profit margin.

Solution:

- Promote high-profit products along with essential items
- Use combo offers
- Increase overall profit without reducing sales

4.11 Weekly Review System

Problem: Problems are identified late due to lack of monitoring.

Solution:

- Check stock levels every week
- Review pending dues
- Track daily sales
- Identify and solve issues quickly

Overall Recommendation Summary:

The implementation of these recommendations will transform the store from a manually managed system to a more structured and data-driven operation. By improving cash flow management, optimizing inventory, and leveraging data analytics, the store can achieve higher profitability, better customer satisfaction, and long-term sustainability.

In the long run, adopting these data-driven practices will enable the store to make informed decisions, reduce financial risks, and build a scalable and efficient business model suitable for rural retail environments.